

Ch. 13 — P02 (K) Nelder-Mead on a Stochastic Response Surface

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Background

Nelder-Mead (simplex search) is a derivative-free optimizer suitable when $g(x)$ is expensive to evaluate and gradients are unavailable or unreliable. `scipy.optimize.minimize(..., method='Nelder-Mead')` wraps this algorithm. When applied to a stochastic objective, each evaluation should use *multiple replications* to reduce noise.

Setup

The following function represents the true (unknown) daily cost of an (s, S) inventory policy as a function of the reorder point s and order-up-to level S :

Tasks

- (a) Evaluate `inventory_cost([20, 80])` and `inventory_cost([40, 100])` with `n_reps=30`. Report the mean and standard deviation across replications for each configuration.
- (b) Apply `scipy.optimize.minimize` with `method='Nelder-Mead'` to minimize `inventory_cost` starting from $x_0 = [30, 90]$. Use `n_reps=10` per evaluation during the search, then re-evaluate the reported optimum with `n_reps=50` for a more precise estimate.
- (c) Plot the optimizer's trajectory in (s, S) space by logging each function evaluation. Use `callback` or wrap the objective to append `(params, cost)` to a list. Scatter-plot the evaluations colored by call order.
- (d) Nelder-Mead is sensitive to initialization. Run the optimizer from three different starting points: $x_0 = [10, 50], [30, 90], [60, 120]$. Do all three converge to the same (s^*, S^*) ? Discuss what this reveals about the response surface.

- (e) **Bias from noisy evaluations:** when $n_{\text{reps}}=10$, the objective is noisy and Nelder-Mead may accept a move to a worse point if the noisy estimate looks better. Propose one mitigation strategy (without switching algorithms) and implement it.

Constraints

$1 \leq s < S \leq 200$, both integers. Round non-integer suggestions from the optimizer before passing to the simulator.